

GLA UNIVERSITY, GREATER NOIDA

Greater Noida – 201310, Uttar Pradesh, India

HEALTH OS: A HEALTHCARE SUPER APPLICATION

SOFTWARE REQUIREMENT SPECIFICATION

Version 1.0

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING - AI

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GLA UNIVERSITY, GREATER NOIDA – 201310

September 2026

Revision History

Version	Date	Change	Reason
1.0	20-09-2026	Faculty-reviewed SRS baseline with architecture, planning, requirements, data, and future-scope updates.	Prepared for academic review and final submission.

1. Introduction

1.1 Purpose

This document specifies the requirements for Health OS v1.0, a healthcare web application that connects patient registration, doctor discovery, appointment management, electronic health records, electronic prescriptions, patient consent, and one AI-assisted medical-image analysis feature. It is the baseline against which the implementation and test cases for this release will be verified.

1.2 Scope

Health OS v1.0 provides a patient and doctor workflow for finding doctors, managing appointment slots, booking consultations, completing visits, creating electronic health records, issuing prescriptions, and controlling access to health information through consent. The release also includes an AI-assisted skin-lesion malignancy classification feature based on a selected medical-image dataset. The AI output is intended to support clinical review and is not presented as a standalone medical diagnosis.

The system does not implement pharmacy orders, laboratory booking and report exchange, hospital admission, insurance claims, ambulance dispatch, home-care booking, payment processing, or ABDM/ABHA interoperability in this release. These remain part of the longer-term Health OS vision.

The broader Health OS vision includes general AI-assisted symptom triage and report summarization; for v1.0, the AI scope is narrowed to one measurable medical-image classification task so that the dataset, baseline, model, and evaluation can be validated within the academic project timeline.

1.3 Definitions

Term	Definition
EHR	Electronic Health Record: the patient's longitudinal digital health record maintained by the system.
Consent record	A stored record showing what health information a patient has authorized a doctor to access, when that authorization was granted, and whether it was revoked.
Availability slot	A future time interval published by a doctor that can be booked by one patient.
AI prediction	The output of the trained image-classification model for an uploaded skin-lesion image; it is decision support, not a confirmed clinical diagnosis.
Patient-level split	A dataset split in which images belonging to the same patient are kept in only one of train, validation, or test sets to reduce leakage.

2. Overall Description

2.1 User Classes

User class	Who they are	What they may do
Patient	Authenticated person seeking healthcare.	Register/login, search doctors, book/reschedule/cancel appointments, view own EHR and prescriptions, upload an image for AI-assisted analysis, and grant/revoke doctor access.
Doctor	Independent or hospital-affiliated practitioner using the platform.	Manage availability slots, view appointments, complete consultations, access records after consent, issue prescriptions, and review AI-assisted prediction results.
Admin	Authorized platform or clinic operations user.	Manage permitted user/account and appointment administration, monitor system activity, and review audit information. Admin accounts are provisioned by an authorized system owner rather than created through public registration; role-based access does not bypass patient consent for protected health records.

2.2 Assumptions and Dependencies

- The application depends on an SMS/WhatsApp/email provider for appointment confirmations and reminders; if the provider is unavailable, the appointment remains stored and the notification attempt is recorded.
- Patients and doctors require a smartphone or computer with internet access. Image upload requires a supported image file and is subject to the configured file-size limit.
- PostgreSQL is the primary relational database. Redis is used for temporary OTP/session data and API rate limiting.
- The AI feature depends on the locally downloaded ISIC training images, metadata, and ground-truth labels. The model is trained offline and the saved model artifact is loaded by the online prediction service.
- ABDM/ABHA integration depends on sandbox access and credentials that are not currently available.

2.3 Constraints

- C1 The project is a single-student academic build with a fixed semester timeline; therefore the v1.0 release covers only a limited set of Health OS modules.
- C2 Health data requires controlled access, consent handling, and audit logging from the first release; these cannot be deferred to a later module.
- C3 The AI feature is limited to one image-classification task for v1.0. Building a general multi-disease medical AI system is outside the schedule and available compute budget.
- C4 The selected dataset is a research dataset; performance on it cannot by itself establish clinical validity for real patients.

2.4 Software Development Methodology

Health OS will be developed using an incremental, Agile-style sprint methodology. Requirements are divided into working increments so that the core appointment and EHR workflow becomes demonstrable before the AI feature is integrated. Each increment follows requirement selection, design, implementation, testing, review, and integration. Requirement changes are recorded through the Revision History rather than silently changing the specification.

3. Data Source and Baseline

The AI feature uses the ISIC 2024 SLICE-3D Permissive dataset. The ISIC Archive lists 217,477 JPEG skin-lesion image crops, 217,477 metadata entries, and 217,477 gold-standard malignancy labels for this release. The dataset is available under CC BY. The project will use the binary malignancy status as the prediction target. Source: ISIC Challenge dataset page and the SLICE-3D data descriptor.

Item	Value	Status
Dataset name	ISIC 2024 SLICE-3D Permissive	Verified from official ISIC source
Source	ISIC Archive; DOI: 10.34970/2024-slice-3d-permissive	Verified
Licence / terms	Creative Commons Attribution (CC BY)	Verified
Images	217,477 JPEG lesion crops	Verified
Metadata entries	217,477	Verified
Ground-truth entries	217,477 gold-standard malignancy values	Verified
Prediction target	Benign vs malignant lesion status	Verified target definition
Task type	Binary image classification	Defined for this release
Class balance	Exact percentages to be calculated from the downloaded ground-truth CSV before training; no unverified percentage is stated here.	Pending local data audit
Primary metric	Recall for the malignant class	Chosen because missing a malignant case is more consequential than missing a benign case; recall measures the proportion of malignant cases that the model detects.
Level 0 baseline	Majority-class prediction; score to be computed from the local ground-truth file before training.	Required before model training
Level 1 baseline and target	To be added after the simple baseline is trained and evaluated.	Later, not invented in v1.0

Data audit note **The exact class balance and Level-0 score are intentionally not fabricated. They will be generated from the downloaded ground-truth CSV and recorded in the next revision once the local dataset is audited. This follows the template requirement to distinguish verified facts from values still being measured.**

Dataset suitability and limitations

The dataset contains de-identified skin-lesion crops with patient identifiers that can be used to keep the same patient out of multiple data splits. Its image labels include a malignant class based on retrospective pathology review and a benign class that combines strong and weak labels. The data descriptor also notes that skin tone was not documented, so the model must not be described as clinically validated across all populations.

Model baseline plan

The first baseline is the Level 0 majority-class predictor. The final model is planned as a transfer-learning image classifier such as EfficientNet-B0, selected for a manageable compute footprint. A patient-level split will be used to reduce leakage. The final release target will be set only after the Level 0 and Level 1 results are measured.

4. System Architecture

Health OS uses a modular monolith: one FastAPI backend with clear module boundaries, rather than independently deployed microservices. The architecture separates offline model training from online prediction so the deployed application loads a versioned model artifact instead of retraining during a patient request. Offline model training is separated from the online inference service; the deployed application loads only the approved model artifact.

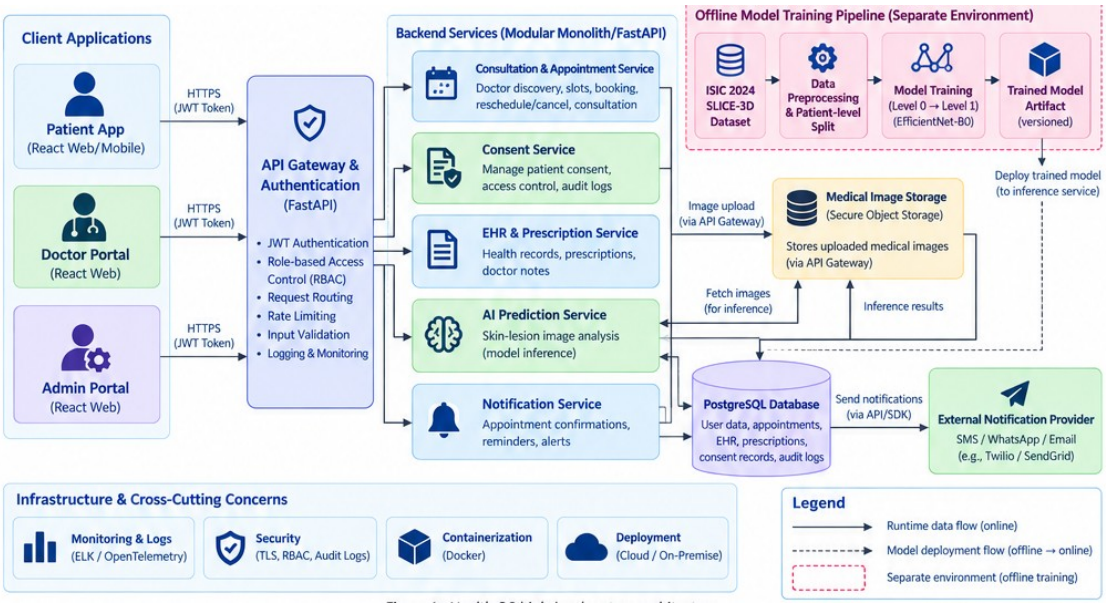


Figure 1 - Health OS high-level system architecture

Component	Responsibility
Patient App (React)	Provides patient registration, appointment actions, health-record access, consent controls, and image upload through the API Gateway.
Doctor Portal (React)	Provides doctor slot management, appointment workflow, authorized EHR access, prescriptions, and AI-result review.
Admin Portal (React)	Provides authorized operational and account-management functions; it does not bypass consent rules for private medical records.
API Gateway & Authentication	Single entry point that validates requests, verifies JWTs, enforces RBAC, routes business requests to the correct module, and forwards validated image uploads to secure medical-image storage.
Consultation & Appointment Service	Handles doctor search, availability, slot creation, booking, rescheduling, cancellation, and appointment completion.
Consent Service	Stores grant/revoke/check decisions and allows the EHR module to enforce patient authorization.

Component	Responsibility
EHR & Prescription Service	Creates health-record entries from completed consultations and stores prescriptions linked to completed appointments.
AI Prediction Service	Accepts a validated image, loads the saved model artifact, performs inference, and returns a prediction and score for doctor review.
Medical Image Storage	Stores validated uploaded images using a generated object key and database reference; all upload and retrieval requests are routed through the API Gateway.
Notification Service	Creates confirmation/reminder messages and calls the configured external notification provider.
PostgreSQL	Primary relational store for users, appointments, records, consent, images, predictions, notifications, and audit logs.
Redis	Stores short-lived OTP/session data and supports API rate limiting.
Offline Model Training Pipeline	Performs dataset preparation, patient-level splitting, baseline evaluation, transfer-learning training, and model versioning before deployment.

5. Functional Requirements

5.1 Authentication and Role-Based Access

FR-1.1 The system shall allow a person to create a Patient or Doctor account using a mobile number verified by OTP.

FR-1.2 The system shall reject an incorrect OTP after three consecutive failures and lock further OTP attempts for that number for 10 minutes.

FR-1.3 The system shall issue a signed JWT after successful authentication and reject expired or invalid JWTs with HTTP 401.

FR-1.4 The system shall restrict each protected API endpoint to the roles permitted for that endpoint.

FR-1.5 The system shall allow an authorized system owner to provision an Admin account; public registration shall not create Admin accounts.

FR-1.6 The system shall allow an Admin to perform only the administrative functions permitted by the assigned role and shall prevent direct access to protected health records without the required authorization.

5.2 Consultation and Appointment Management

FR-2.1 The system shall allow a patient to search doctors by specialization and display matching doctor profiles.

FR-2.2 The system shall allow a doctor to create future availability slots with a start time, end time, and appointment duration.

FR-2.3 The system shall display the current available slots for a selected doctor.

FR-2.4 The system shall allow a patient to book an available slot.

FR-2.5 The system shall prevent two patients from creating active appointments for the same doctor and slot.

FR-2.6 The system shall allow a patient to reschedule a booked appointment up to one hour before its scheduled time.

FR-2.7 The system shall allow a patient to cancel a booked appointment up to one hour before its scheduled time.

FR-2.8 The system shall allow an appointment to be recorded as either video or in-person.

FR-2.9 The system shall allow a doctor to mark an attended appointment as completed.

5.3 EHR, Prescriptions and Consent

FR-3.1 The system shall create a health-record entry when a doctor completes an appointment.

FR-3.2 The system shall allow a patient to view their own health records and prescriptions.

FR-3.3 The system shall allow a patient to grant a doctor access to specified health-record data.

FR-3.4 The system shall allow a patient to revoke an active consent record.

FR-3.5 The system shall deny a doctor access to a patient's protected health record when no valid consent or permitted access rule exists.

FR-3.6 The system shall allow a doctor to create an e-prescription linked to a completed appointment.

5.4 Model-Backed Requirements - AI Medical Image Prediction

FR-4.1 The system shall accept an uploaded image in a supported image format and reject unsupported formats or files that exceed the configured size limit with a validation error.

FR-4.2 The system shall send a validated image to the AI prediction service and return the predicted malignancy class for a successful request.

FR-4.3 The system shall present the model score or confidence together with the predicted class and a statement that the result requires clinical review.

FR-4.4 The system shall display "Prediction unavailable" when the prediction service is unavailable or the model returns no usable output.

FR-4.5 The system shall allow a doctor to record a review note or final clinical decision without modifying the original model prediction.

FR-4.6 The system shall not automatically diagnose, prescribe treatment, or deny an appointment based only on the AI prediction.

5.5 Notifications

FR-5.1 The system shall create a notification request after a successful appointment booking.

FR-5.2 The system shall record the delivery status returned by the configured SMS, WhatsApp, or email provider.

FR-5.3 The system shall generate a reminder notification for an upcoming appointment according to the configured reminder schedule.

6. Non-Functional Requirements

NFR-1 The system shall return doctor-search and appointment-booking responses within 2 seconds at the 95th percentile under a test load of up to 50 concurrent users.

NFR-2 The deployed API shall enforce JWT-based authentication and role-based authorization, and health-data traffic shall use TLS 1.2 or higher.

NFR-3 For each AI prediction the system shall record the image ID, prediction ID, model version, timestamp, predicted class, and model score for later evaluation.

NFR-4 The deployed system shall maintain at least 99% monthly availability for appointment and record access, excluding scheduled maintenance windows announced at least 24 hours in advance.

7. Project Plan (Phase-wise)

7.1 Minimum Viable Product

The MVP is the smallest working healthcare workflow that can be demonstrated without depending on the AI model. The project then proceeds through research, design, core development, AI model development, testing, and deployment phases.

FR / NFR ID	Requirement (short)	Why this one is in the MVP
FR-1.1	Register/login via OTP	Every protected feature needs an authenticated user; without this, the system cannot demonstrate role-based flows.
FR-2.1	Search doctors by specialization	It is the first step in the patient journey and establishes the input to appointment booking.
FR-2.2	Doctor creates availability slots	Patients cannot book reliably unless the doctor-controlled availability is represented in the system.
FR-2.4	Book an available appointment	This is the core user action that turns the system from a directory into an appointment platform.
FR-2.5	Prevent double booking	A booking workflow is not usable if two users can reserve the same slot.
NFR-1	2-second booking response at 95th percentile	Response time can be demonstrated objectively during testing and is visible in a live booking flow.
NFR-2	JWT/RBAC and TLS protection	Protected healthcare workflows require authenticated, role-limited API access and secure transport from the first demonstrable release.

Deliberately not in the MVP **FR-4.2 AI prediction. Before adding it to the release, the project must complete the dataset audit, Level-0 baseline, a simple Level-1 baseline, model evaluation, and inference integration.**

7.2 Phase-wise Development Plan

Phase	Period	Key Activities	Major Deliverables
Phase 1	1 Aug – 1 Sep 2026	Research & Requirement Analysis: study the healthcare domain, review literature, inspect the selected ISIC dataset, define requirements, and finalize the technology stack.	SRS, problem statement, dataset analysis report.
Phase 2	2 Sep – 16 Sep 2026	System Design: finalize high-level architecture, database/ER design, UI/UX flow, API contracts, consent flow, and security	HLD, ER diagram, UI prototypes, API design.

Phase	Period	Key Activities	Major Deliverables
		controls.	
Phase 3	17 Sep – 17 Oct 2026	Core Development (MVP): authentication, roles, doctor discovery, slot management, appointment booking, EHR, and core consent workflow.	Working web application with core healthcare workflow.
Phase 4	18 Oct – 7 Nov 2026	AI Model Development: data preprocessing, patient-level split, Level-0/Level-1 baselines, transfer-learning training, evaluation, and inference integration.	Trained model artifact, inference API, evaluation results.
Phase 5	8 Nov – 17 Nov 2026	Testing & Validation: functional testing, integration testing, security checks, dataset/model validation, and defect correction.	Test report, bug log, measured model performance.
Phase 6	18 Nov – 30 Nov 2026	Deployment & Documentation: Docker deployment, final bug fixes, user guide, technical documentation, and final report preparation.	Deployed system, documentation, final submission.
Final Presentation	5 – 10 Dec 2026	Demonstration, report submission, presentation, and viva preparation.	Final presentation and viva.

8. Data Requirements

8.1 Entity: Appointment

Field	Type	Constraint / Notes
id	UUID	Primary key.
patient_id	UUID	Foreign key -> Patient; not null.
doctor_id	UUID	Foreign key -> Doctor; not null.
slot_id	UUID	Foreign key -> AvailabilitySlot; not null; identifies the booked slot.
scheduled_at	datetime	Not null; must be in the future when the booking is created.
duration_minutes	integer	Positive value; determines the slot length.

Field	Type	Constraint / Notes
mode	string	One of video or in_person.
status	string	booked, completed, cancelled, or no_show; defaults to booked.
created_at	datetime	Automatically generated; not null.

Other entities and purpose

Entity	Purpose
User	Stores common identity and authentication data shared by platform roles.
Patient	Stores patient-specific profile data linked one-to-one with User.
Doctor	Stores specialization, license number, HPR field, and doctor-specific profile data.
Admin	Stores administrative profile data linked to a User account.
AvailabilitySlot	Stores doctor-created future time slots offered for booking.
HealthRecord	Stores consultation summaries and structured record entries associated with a patient and completed appointment.
Prescription	Stores prescriptions issued by doctors for completed appointments.
ConsentRecord	Stores patient grant/revoke decisions for doctor access to protected records.
MedicalImage	Stores the application reference for a patient-uploaded image and its storage location.
AIPrediction	Stores the prediction class, score, model version, timestamp, and the doctor's review note without changing the original prediction.
Notification	Stores notification requests and provider delivery status.
AuditLog	Stores important security and record-access events with actor and timestamp.

8.2 Entity Relationship Diagram

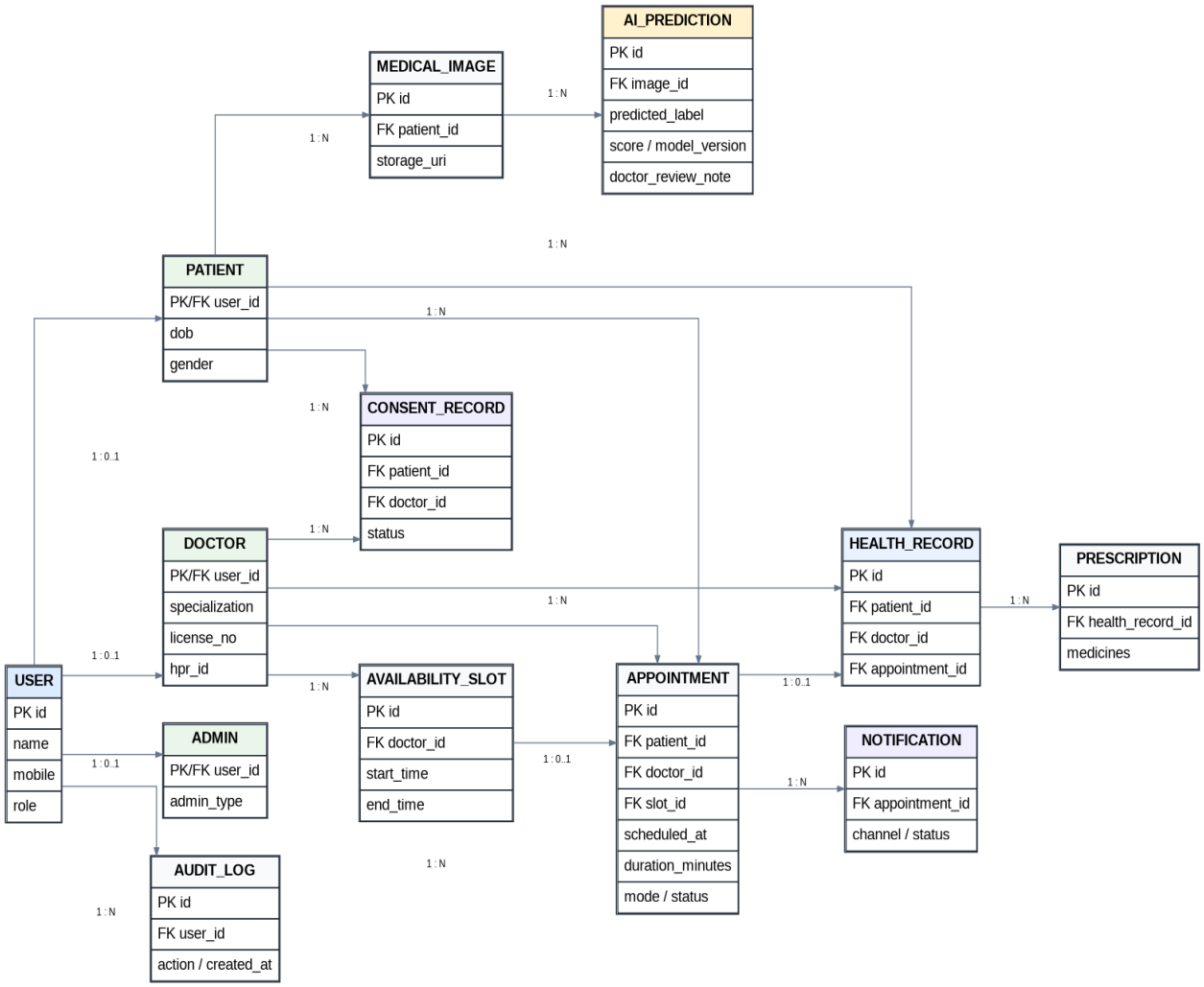


Figure 2 - Health OS v1.0 entity relationships and cardinalities

The ER model keeps User as the common identity entity and separates role-specific profiles for Patient, Doctor, and Admin. Appointment links a patient and doctor to one AvailabilitySlot; completed appointments may create a health record. Consent records govern doctor access; medical images can generate AI prediction records; AI predictions retain a doctor review note; notifications are linked to appointments; audit logs provide an accountability trail.

9. Design Decisions and Alternatives Considered

#	Decision	Alternative rejected	Reason
D1	Modular monolith using FastAPI with folder-level module boundaries.	Microservices from the first release.	A single-student project has a fixed academic timeline. One deployable backend reduces deployment and networking overhead while preserving clear module boundaries for future separation.

#	Decision	Alternative rejected	Reason
D2	FastAPI + PostgreSQL + Redis.	Django + PostgreSQL as the primary backend stack.	FastAPI provides typed request validation and OpenAPI documentation, while PostgreSQL provides relational constraints and transaction support needed for booking, consent, and EHR data. Redis is limited to short-lived OTP/session/rate-limit data.
D3	Transfer-learning image classifier with a patient-level split.	Training a large neural network from scratch or using an image-level random split.	The project has limited compute and must avoid leakage between related images from the same patient. Transfer learning is practical for the available hardware, while patient-level splitting better reflects the intended evaluation setup.

10. Acceptance Criteria

ID	Live demonstration	Expected outcome
AC-1	Create a new patient account with a valid mobile number and valid OTP.	The patient is authenticated and a valid session/JWT is returned.
AC-2	Search a specialization, select a doctor slot, and confirm an appointment.	The appointment appears under the patient's appointments and the slot is no longer available.
AC-3	Use two sessions to book the same doctor slot within the same test window.	Exactly one booking succeeds; the second request receives a slot-unavailable error and creates no duplicate appointment.
AC-4	Complete the booked appointment from the doctor portal.	A health-record entry is created for the patient. The patient can view it according to the configured authorization rule.

11. Future Scope

- Integration with pharmacy services for medicine ordering and home delivery.
- Laboratory booking and exchange of laboratory reports.
- Hospital admission, bed management, and broader hospital workflows.
- Insurance policy, claims, and cashless healthcare workflows.
- Ambulance dispatch and emergency-care integration.
- ABDM/ABHA interoperability and secure health-record exchange when required credentials are available.
- Telemedicine, home-care, and remote patient monitoring.
- Expanded medical-AI capabilities such as symptom triage, report summarization, and additional disease/image models.
- Android/iOS mobile applications with offline capabilities and notification support.
- Wearable and IoT integration for continuous health monitoring and analytics.